Semester Test 1

Electricity and Electronics EBN 122

23 August 2019

Test Information			
Maximum marks:	45	Full marks:	45
Duration of test:	90 min + (5 min before and 10 min after for OCR)	Open/Closed book:	Closed
Total number of pages (this page included)		18	

Important

1. The examination regulations of the University of Pretoria apply.
2. Answer all questions on the OCR forms provided. The question number **01** in this question paper refers to row 01 of the OCR form.
3. Read the instructions on the OCR form carefully.
4. Where applicable, answers must include units.
5. Final answers must be written as real numbers and not as fractions.
6. Use at least 3 significant numbers to write irrational numbers.

Lecturers: Prof J.P. Jacobs, Prof J.P. de Villiers, Dr X. Ye

INSTRUCTIONS

Do step 0 in the first 5 minutes of the session. (-5 to 0 min)

Do step 1 in the next 90 minutes of the test. (0-90 min)

Do step 2 within the next 10 minutes. (90-100 min)

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form, and also fill in your name and student number on additional pages of the OCR form.

STEP 1 (0 min to 90 min): Do your own calculations on the question paper. Write down the answers for numeric questions on the PRACTICE ANSWER SHEET. Be sure to include the units where applicable.

STEP 2 (90 min to 100 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

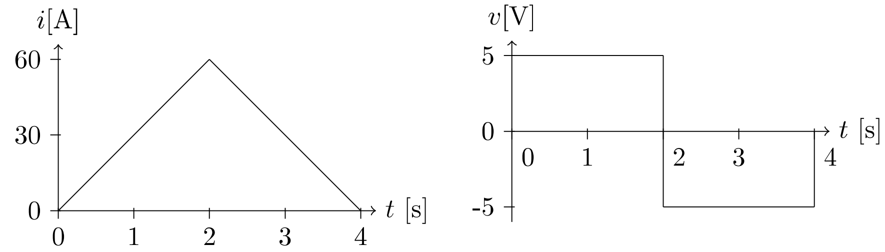
- The Assessment ID is: **2019EBN122S01**
- Use a mechanical pencil with 0.5 mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops as decimal points.

Section 1

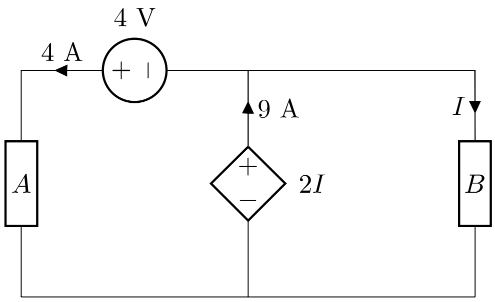
Questions 1 to 3 [3]	
A lightning bolt induces a constant current of 10 kA for 15 μs as it strikes the ground.	
01	How much charge Q_l is transferred by the lightning bolt? [1]
02	If the voltage of the bolt is 10^9 V with reference to the ground, how much power P_l is dissipated by the lightning bolt during the strike? [1]
03	Calculate the total energy E_l of the lightning bolt in Wh. [1]

Question 4 [2]

The figure below shows the current through and the voltage across a circuit element.



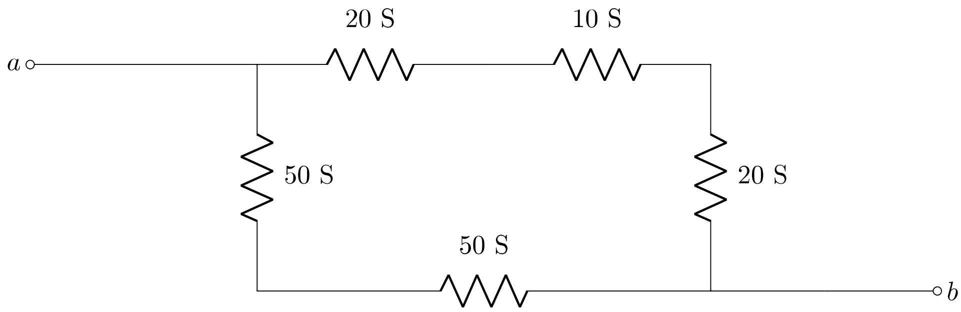
04	Find the total energy E_e absorbed by the element for the interval $0 \leq t \leq 4$ s. [2]

Questions 5 to 7 [3]	
Calculate the power relating to the following elements in the circuit below:	
	
05	Calculate the power P_D associated with the dependent voltage source. [1]
06	Is the dependent voltage source in the circuit supplying power (write a 1) or absorbing power (write a 2)? [1]
07	Calculate the resistance R_B of circuit element B. [1]

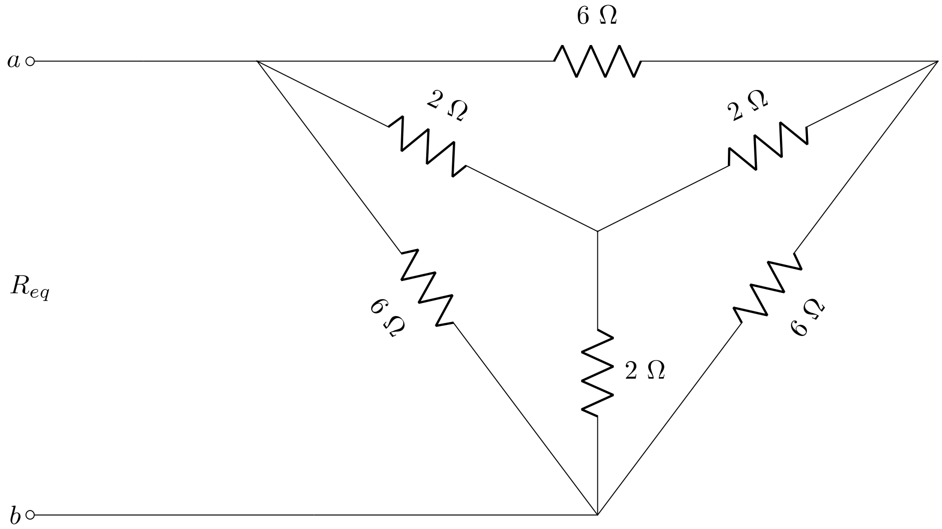
Total Section 1: [8]

Section 2

Question 8 [2]

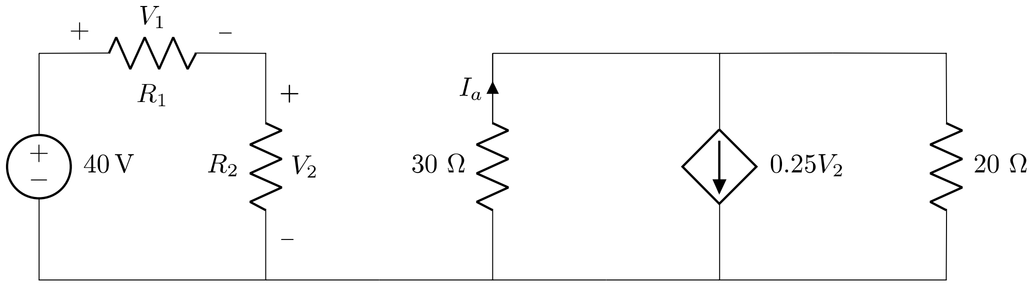


08	Determine the equivalent conductance G_{ab} between nodes a and b. [2]

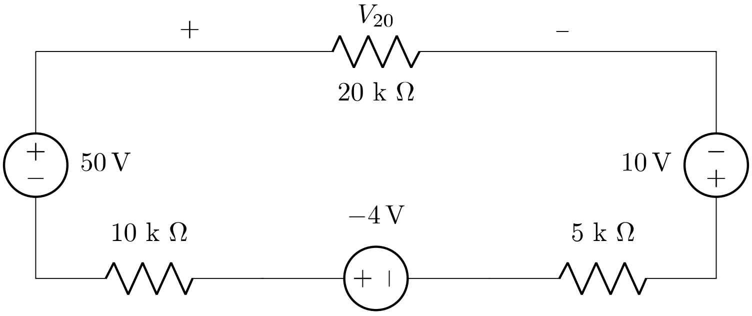
Question 9 [2]	
	
09	Determine the equivalent resistance R_{eq} between nodes a and b. [2]

Questions 10 to 11 [4]

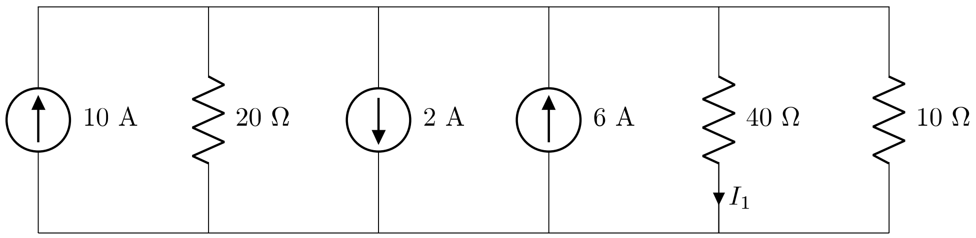
Consider the circuit below and assume that $V_1/V_2 = 3$ and $R_2 = 20\ \Omega$.



10	Calculate R_1. [2]
11	Calculate I_a. [2]

Question 12 [2]	
	
12	Determine V_{20}. [2]

Question 13 [2]



13	Determine I_1. [2]

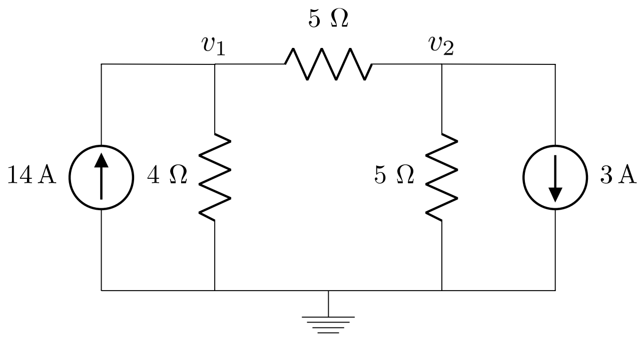
Total Section 2: [12]

Section 3

Question 14 [2]	
14	Determine θ if $v_1 = -2$. [2] $$\begin{bmatrix} -2\theta & -4 \\ 6 & 4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -2\theta \\ 12 \end{bmatrix}$$

Question 15 [2]	
15	Determine σ if $i_3 = -1.2$. [2] $$\begin{bmatrix} 2 & 1 & 4 \\ 0 & -2 & 1 \\ 2 & 4 & 0 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} \sigma \\ 4 \\ -2 \end{bmatrix}$$

Questions 16 to 19 [4]
Use nodal analysis to set up the following two equations of the form: $$av_1 + bv_2 = 280$$ $$cv_1 + dv_2 = 15$$ Calculate the unknown coefficients a, b, c and d. Note: Coefficients should be integer numbers.



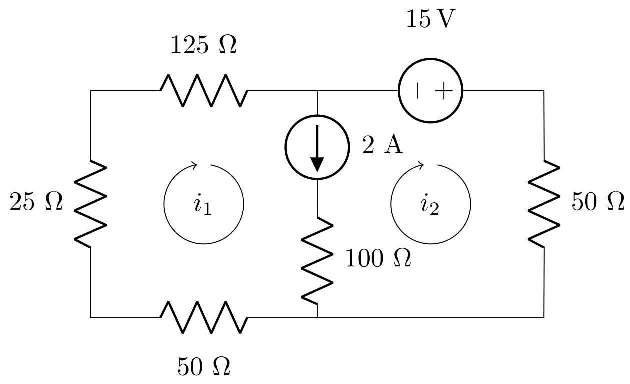
16	a [1]	**17**	b [1]	**18**	c [1]	**19**	d [1]

Questions 20 to 23 [4]

Use mesh analysis to set up the following two equations of the form:

$$ei_1 + fi_2 = 15$$
$$gi_1 + hi_2 = 2$$

Calculate the unknown coefficients e, f, g and h. Note: Coefficients should be integer numbers.



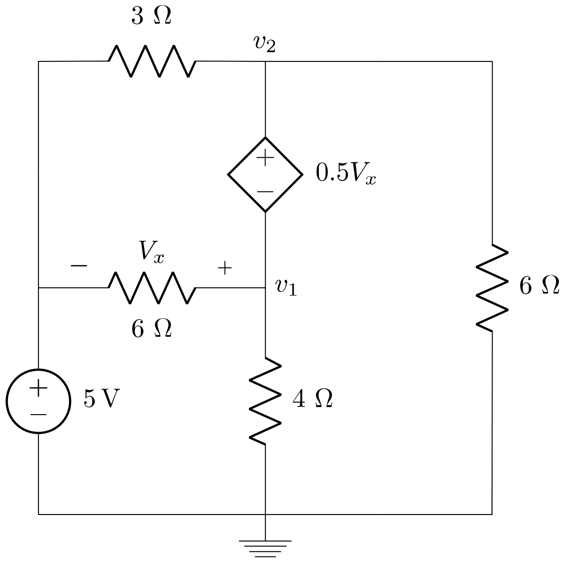
20	e [1]	**21**	f [1]	**22**	g [1]	**23**	h [1]

Questions 24 to 27 [4]

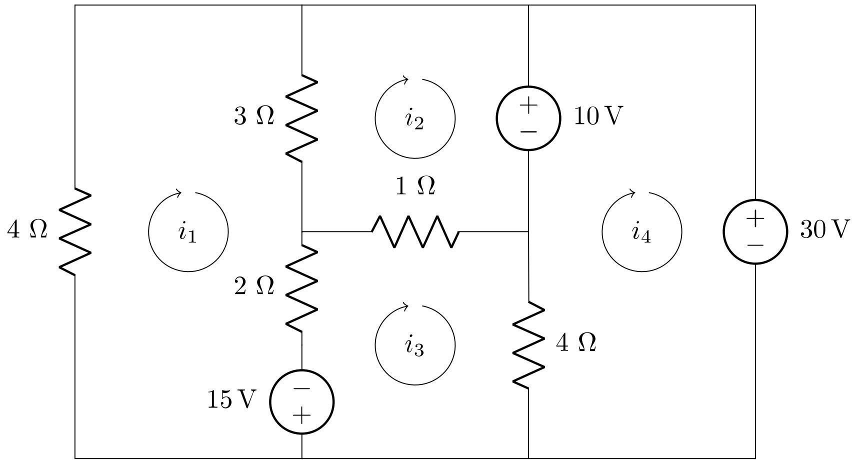
Use nodal analysis to set up the following two equations of the form:

$$qv_1 + rv_2 = 30$$
$$13v_1 + sv_2 = t$$

Calculate the unknown coefficients q, r, s and t. Note: Coefficients should be integer numbers.

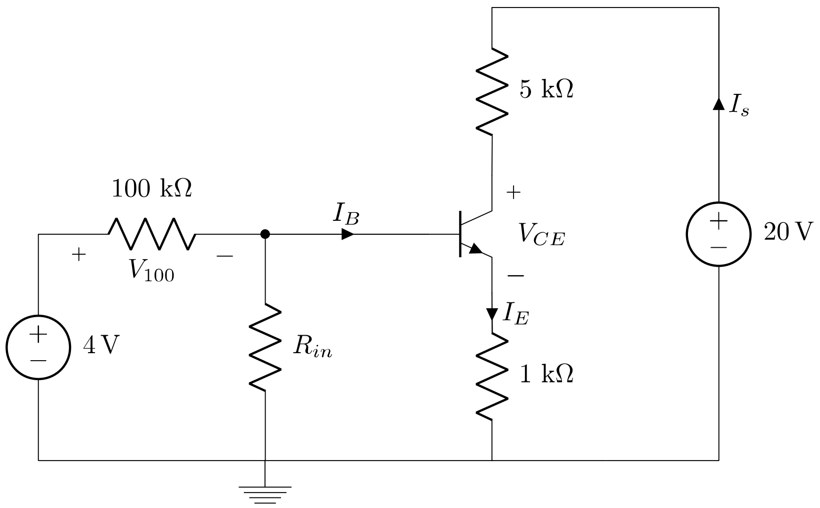


24	q [1]	**25**	r [1]	**26**	s [1]	**27**	t [1]

Questions 28 to 31 [4]
Use inspection to compile a matrix equation of the form below $$\begin{bmatrix} 9 & -3 & -2 & 0 \\ -3 & v & -1 & 0 \\ -2 & -1 & 7 & w \\ 0 & 0 & w & 4 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \end{bmatrix} = \begin{bmatrix} 15 \\ x \\ -15 \\ y \end{bmatrix}$$ and provide the values of the matrix elements v, w, x and y.


28	v [1]	**29**	w [1]	**30**	x [1]	**31**	y [1]

Questions 32 to 35 [5]
Consider the case where $I_E = 1.01$ mA and $\beta = 100$ in the circuit below. Let $V_{BE} = 0.7V$, i.e. the transistor is operating in the active region. Determine I_B, V_{100}, I_s and V_{CE}.



32	I_B [1]	**33**	V_{100} [2]	**34**	I_s [1]	**35**	V_{CE} [1]

This page is intentionally left open for rough work.

Total Section 3: [25]

Grand Total : [45]

PRACTICE ANSWER SHEET

Assessment ID	2	0	1	9	E	B	N	1	2	2	S	0	1

	Answer	Unit
01	$Q_l =$	
02	$P_l =$	
03	$E_l =$	
04	$E_e =$	
05	$P_D =$	
06	1 or 2?	No units
07	$R_B =$	
08	$G_{ab} =$	
09	$R_{eq} =$	
10	$R_1 =$	
11	$I_a =$	
12	$V_{20} =$	
13	$I_1 =$	
14	$\theta =$	No units
15	$\sigma =$	No units
16	$a =$	No units
17	$b =$	No units
18	$c =$	No units
19	$d =$	No units
20	$e =$	No units
21	$f =$	No units
22	$g =$	No units
23	$h =$	No units
24	$q =$	No units
25	$r =$	No units
26	$s =$	No units
27	$t =$	No units
28	$v =$	No units
29	$w =$	No units
30	$x =$	No units
31	$y =$	No units
32	$I_B =$	
33	$V_{100} =$	
34	$I_s =$	
35	$V_{CE} =$	